
MODEL SELECTION AND VALIDATION REPORT (general_teve model)

OBSERVATIONS:3382

SANITY: OK - all models in the selection process passed sanity checks.

SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

MODEL SUMMARY (general_teve model)

Spatiotemporal model fit by REML ['sdmTMB']

Formula: fixed_formula

Time column: year

Family: gaussian(link = 'identity')

Conditional model:

	coef.est	coef.se
(Intercept)	-0.01	0.07
sbt_mean	0.14	0.04
sbt_sd	-0.01	0.03
sst_sd	-0.01	0.03
sic_mean	-0.08	0.04
log_chla_mean	-0.02	0.02
depth	0.17	0.03

Dispersion parameter: 0.83

Spatiotemporal AR1 correlation (rho): 0.82

Matérn range: 165.74

Spatiotemporal marginal AR1 SD: 0.55

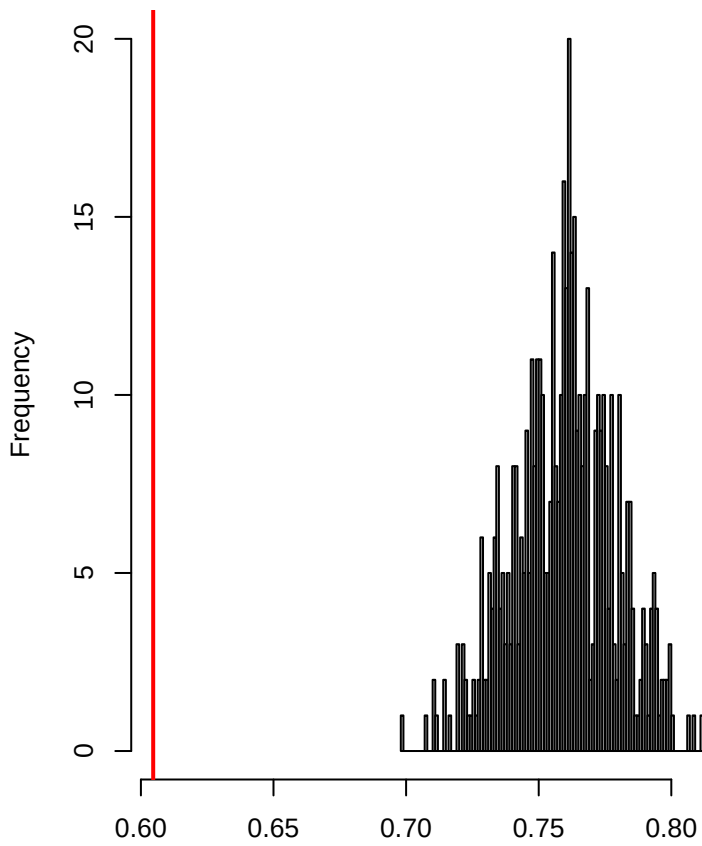
REML criterion at convergence: 4449.663

See ?tidy.sdmTMB to extract these values as a data frame.

SELECTION STATISTICS

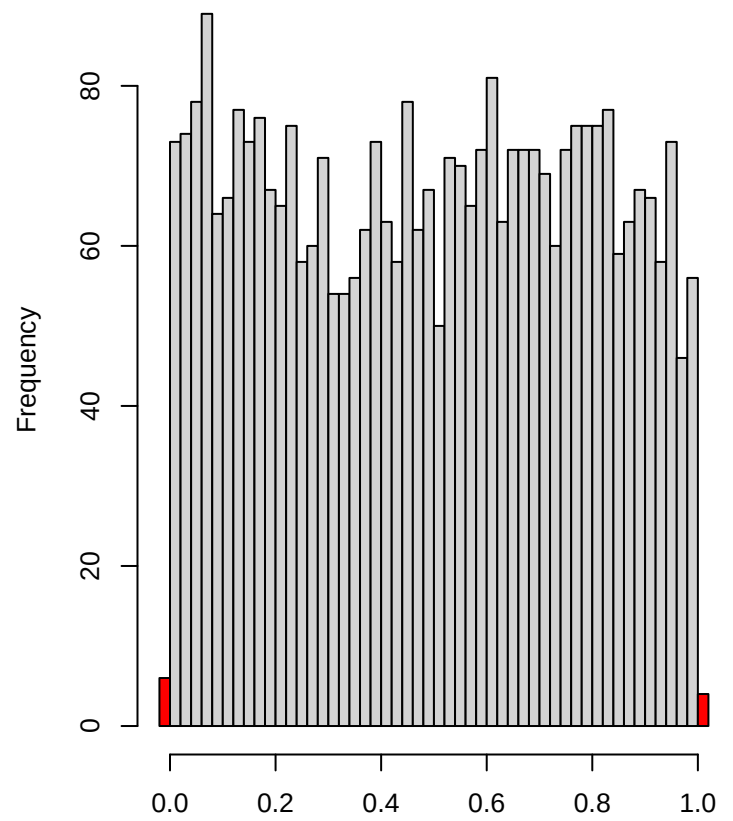
	df	AIC	delta_AIC
spatiotemporal_random_fields_ar1	11	8921.325	2.457658
spatiotemporal_random_fields_ar1_with_anisotropy	13	8923.783	162.114466
spatiotemporal_random_fields_iid	10	9085.898	3.738150
spatiotemporal_random_fields_iid_with_anisotropy	12	9089.636	39.436950
spatial_random_fields	10	9129.073	3.656533
spatial_random_fields_with_anisotropy	12	9132.729	201.156981
error_only	8	9333.886	NA

DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated



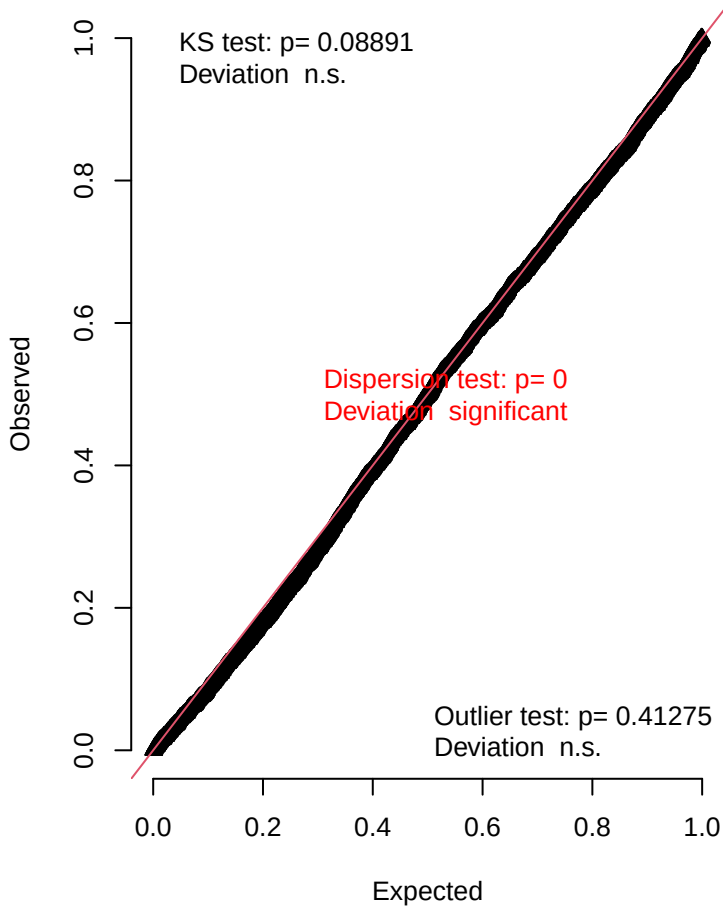
Simulated values, red line = fitted model. p-value (two.sided)

Outlier test n.s.

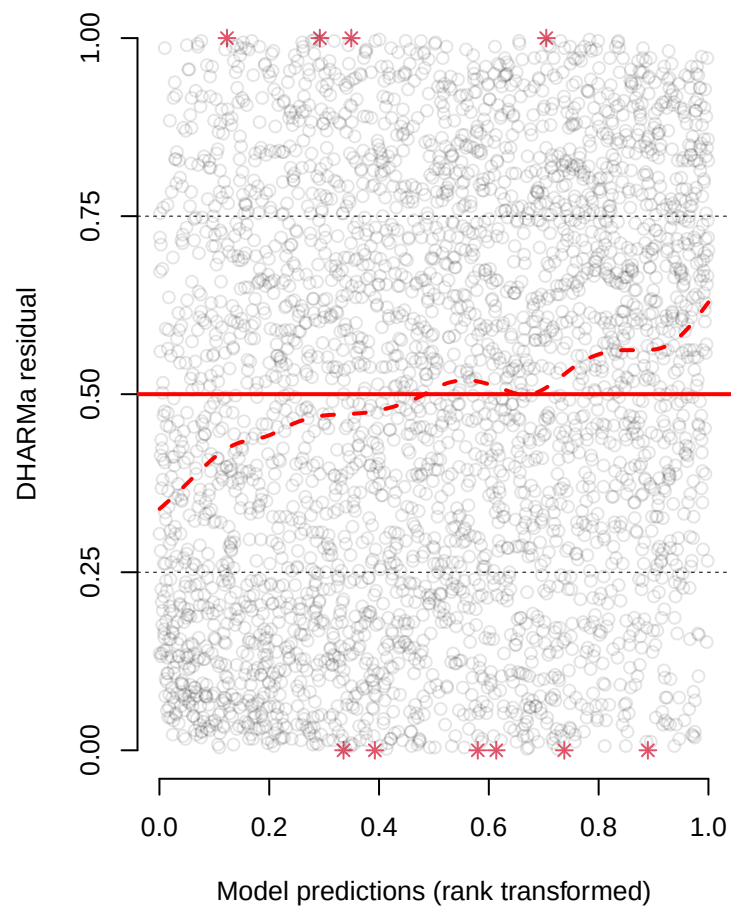


Residuals (outliers are marked red)

QQ plot residuals



DHARMa residual vs. predicted



DHARMA Moran's I test for distance-based autocorrelation
Moran's I = 0.013 | H1: Distance-based autocorrelation | p-value = 0.002

